Ts. Dr. MOHD ZAMRI BIN ZAHIR AHMAD

Senior Lecturer | Professional Technologist (MBOT)

Faculty of Electronic Engineering & Technology (FKTEN), Universiti Malaysia Perlis (UniMAP), Pauh Putra Campus, 02600 Arau, Perlis, Malaysia

Mobile: 011-2882 1198 **|** **Email:** zamrizahir@unimap.edu.my **|** **Nationality:** Malaysian

Scopus: 57192404100 **|** **ORCID:** 0000-0003-3839-2284

PROFILE

Senior Lecturer at the Faculty of Electronic Engineering & Technology, UniMAP (academic staff since 2016), and a registered Professional Technologist (MBOT). Research focuses on metaheuristic optimization for combinatorial (t-way) software testing, fuzzy logic and IoT-based intelligent systems. Principal investigator of an FRGS grant, holder of multiple MyIPO copyright notifications, and recipient of numerous gold medals at national and international innovation exhibitions. Previously an IBM Hardware Engineer supporting enterprise i-Series (AS/400) systems.

Research Interests: Optimization Algorithms, Combinatorial Testing, Software Testing, Fuzzy Logic, IoT Systems

EDUCATION

Doctor of Philosophy (PhD) in Computer Engineering **2023**
Universiti Malaysia Perlis (UniMAP)
Thesis: A Self-Adapting Ant Colony Optimization using Fuzzy Logic (ACOF) for Combinatorial Test Suite Generation

Master of Science (MSc) in Computer Engineering **2016**
Universiti Malaysia Perlis (UniMAP)
Thesis: A T-Way Test Suite Generation Strategy for Sequence-Based Interaction Testing

Bachelor of Engineering (Hons) in Computer Engineering **2012**
Universiti Malaysia Perlis (UniMAP)

Diploma in Computer Engineering **2009**
Universiti Malaysia Perlis (UniMAP)

PROFESSIONAL MEMBERSHIP

Professional Technologist (Ts.), Malaysia Board of Technologists (MBOT) **2024**
Reg. No.: PT24100248 | Field: Electrical & Electronics Technology (EE)

Graduate Technologist, Malaysia Board of Technologists (MBOT) **2021**
Reg. No.: GT21050921 | Field: Electrical & Electronics Technology (EE)

Graduate Engineer, Board of Engineers Malaysia (BEM) **2018**
Reg. No.: G1159102A | Branch: Computer

Member, Persatuan Kakitangan Akademik (PKA) UniMAP **2025 – 2026**
Membership No.: PKA-00503

EMPLOYMENT HISTORY

Senior Lecturer **2023 – Present**
Faculty of Electronic Engineering & Technology (FKTEN), Universiti Malaysia Perlis (UniMAP)

University Lecturer **Sep 2016 – 2023**
Faculty of Electronic Engineering & Technology (FKTEN), UniMAP (2020 – 2023); formerly School of Computer & Communication Engineering (SCCE), UniMAP (2016 – 2020)
Key responsibilities (2016 – Present):

- Teach core undergraduate courses including Computer Architecture, Design Project, Instrumentation, Digital Systems, C Programming, Object-Oriented Programming, Software Engineering and Database Systems.
- Supervise undergraduate and postgraduate research projects in embedded systems, IoT and artificial intelligence applications.

- Coordinate courses and laboratory activities, assessment and academic quality assurance; develop curriculum aligned with OBE and MQF standards.
- Conduct research in optimization algorithms, smart system integration and fuzzy logic control; publish in indexed journals and international conferences.
- Mentor students in innovation, product development and final-year design projects.
- Contribute to faculty management, grant applications and collaborative research initiatives.

Assistant Engineer (IBM Hardware Engineer) **Nov 2012 – Apr 2014**

ALL-Media Integration Technology Sdn. Bhd. (outsourced to IBM Malaysia)

- Hardware engineer for the IBM i-Series (Power Systems / AS/400) platform.
- Resident Engineer for the CIMB 1P project, migrating from mainframe servers to IBM i-Series (Power Systems).
- Performed quarterly preventive maintenance (PM) and software health checks (SHC) on customer i-Series servers.
- Resolved customer hardware failures including storage devices, Hardware Management Console (HMC), I/O cards, optical and tape libraries, and power supplies.

RESEARCH GRANTS

Fundamental Research Grant Scheme (FRGS) **2024**

A New Phase Transition Controller in Sand Cat Swarm Optimization for Combinatorial Test Suite Generation

Role: **Main Researcher** | Amount: RM 124,150 | Duration: 3 years

Industrial Matching Program Grant (IMaP) **2024**

Design and Development of Testing Tool for Solar Harvesting Application

Role: Co-Researcher | Amount: RM 100,000 | Duration: 1 year 6 months

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Notification No.	Year	Title of Work
CRLY2026E09159	2026	Smart IoT-Based Production Output and Defective Product Detection System
CRLY2026E05318	2026	Design and Development of T-Way Test Suite Generator Adopting Sewing Training-Based Optimization Algorithm
CRLY2026E02675	2026	SandCatGo: An Intelligent Online Tool for Combinatorial T-Way Test Suite Generation Based on Sand Cat Swarm Optimization
CRLY2026E02496	2026	Self-Sustained Active Water-Cooling System for PV Modules Using Irradiance-Based Control
CRLY2026E02231	2026	Fuzzy-MedCare: Fuzzy Logic-Based Health Monitoring and Emergency Detection for Home Bedridden Patients
CRLY2025P00907	2025	Intelligent Parcel Delivery and Notification System for Apartment House
CRLY2023P03962	2023	A Self-Adapting Ant Colony Optimization Algorithm using Fuzzy Logic (ACOF) for Combinatorial Test Suite Generation

PUBLICATIONS

Journal Articles

1. Muhammad Aiman, M. A., **Ahmad, M. Z. Z.**, Othman, R. R., Halim, A. A. A., Go, K., Ramli, N., Ahmad, R. B., Kamarudin, L. M., & Al-Walidi, M. M. H. S. (2026). Implementation of sand cat swarm optimization for uniform t-way test suite generation. *Journal of Automation, Mobile Robotics and Intelligent Systems (JAMRIS)*.
2. Abdul Rahim, M. S., Tajuddin, M. F. N., Saad, M. S., Hasanuzzaman, M., **Ahmad, M. Z. Z.**, Halim, A. A. A., & Mid, E. C. (2025). Experimental investigation on improving photovoltaic module efficiency using adaptive fuzzy-PID water cooling. *Solar Energy, 283*, 113828. https://doi.org/10.1016/j.solener.2025.113828
3. Abdul Halim, A. A., Veeraperumal, V., Andrew, A. M., Mohd Yasin, M. N., **Zahir Ahmad, M. Z.**, Hossain, K., Bari, B. S., & Kamal, F. (2023). UWB-based early breast cancer existence prediction using artificial intelligence for large data set. *Journal of Advanced Research in Applied Sciences and Engineering Technology, 29*(2), 81–90.

4. Nasrudin, M. W., Zainuddin, N. S. A., Raja Ahmad, R. A., Yob, R. C., **Ahmad, M. Z. Z.**, et al. (2023). Smart management waiting system for outpatient clinic. *Journal of Advanced Research in Applied Sciences and Engineering Technology, 29*(3), 4861–4870. https://doi.org/10.37934/araset.29.3.4861
5. **Ahmad, M. Z. Z.**, Othman, R. R., & Rashid Ali, M. S. A. (2016). Sequence covering array generator (SCAT) for sequence-based combinatorial testing. *International Journal of Applied Engineering Research, 11*(8), 5984–5991.

Conference Proceedings

1. Muhammad Aiman, M. A., **Ahmad, M. Z. Z.**, Othman, R. R., Halim, A. A. A., Go, K., Al-Walidi, M. M. H. S., Suali, A. J., Rose, N. H. C., & Kamal, F. B. (2026). Variable strength sand cat swarm optimization (VSCSO) for t-way combinatorial test suite generation. In *Proceedings of the International Exchange and Innovation Conference on Engineering & Sciences (IEICES 2025)*.
2. Murad, M. M. H. S., Halim, A. A. A., **Ahmad, M. Z. Z.**, Ramli, N., Zakaria, H. L., Aiman, M. A. M., Go, K., & Rahim, M. S. A. (2026). Application of sewing training-based optimization in generating effective input-output relation-based t-way test suite generation strategies. In *Proceedings of the International Exchange and Innovation Conference on Engineering & Sciences (IEICES 2025)*.
3. **Ahmad, M. Z. Z.**, Hamdi, M. H., Halim, A. A. A., Nasrudin, M. W., Othman, R. R., Raja Ahmad, R. A., & Mat Salleh, S. (2025). IoT warning system for early forest fire detection. In *Advances in Artificial Intelligence and Electronic Design Technologies (ICED 2024)*, Studies in Computational Intelligence, Vol. 1015 (pp. 409–420). Springer. https://doi.org/10.1007/978-3-031-78918-2_36
4. Fong, P. Y., Raja Ahmad, R. A., Ahmad, M. I., & **Ahmad, M. Z. Z.** (2025). IoT based fertilizer system for crop cultivation. In *Advances in Artificial Intelligence and Electronic Design Technologies (ICED 2024)*, Studies in Computational Intelligence (pp. 517–530). Springer. https://doi.org/10.1007/978-3-031-78918-2_45
5. Anvar, H. H., Salleh, S. M., Kamarudin, L. M., & **Ahmad, M. Z. Z.** (2024). Polyvinyl chloride (PVC) ultra-wideband wearable antenna under bending and wet conditions. In *2024 IEEE 1st International Conference on Communication Engineering and Emerging Technologies (ICoCET)* (pp. 1–5). https://doi.org/10.1109/ICoCET63343.2024.10730245
6. **Zahir Ahmad, M. Z.**, Othman, R. R., Ramli, N., & Rashid Ali, M. S. A. (2022). VSTACO: A tuned version of ant colony optimization for generating variable strength interaction in t-way testing strategy. In *International Conference on Software and Computer Applications (ICSCA 2022)*.
7. **Ahmad, M. Z. Z.**, Othman, R. R., Ali, M. S. A. R., Ramli, N., Nasrudin, M. W., & Halim, A. A. A. (2021). A tuned version of ant colony optimization algorithm (TACO) for uniform strength t-way test suite generator: An execution's time comparison. *Journal of Physics: Conference Series, 1962*(1).
8. Ramli, N., et al. (2021). Adopting ant colony optimization algorithm for pairwise t-way test suite generation strategy. *Journal of Physics: Conference Series, 1962*(1).
9. Nasrudin, M. W., Yaakob, S. N., Abdul Rahim, A. N., **Ahmad, M. Z. Z.**, & Ali, M. S. A. R. (2021). Moment invariants technique for image analysis and its applications: A review. *Journal of Physics: Conference Series, 1962*(1).
10. Rashid Ali, M. S. A., Othman, R. R., Yahya, Z. R., & **Ahmad, M. Z. Z.** (2020). A modified artificial bee colony based test suite generation strategy for uniform t-way testing. *IOP Conference Series: Materials Science and Engineering, 767*(1).
11. Ramli, N., Othman, R. R., Khalib, Z. I. A., **Ahmad, M. Z. Z.**, & Fauzi, S. S. M. (2020). Ant colony algorithm to generate t-way test suite with constraints. *Journal of Physics: Conference Series, 1529*(4).
12. **Ahmad, M. Z. Z.**, Othman, R. R., Rashid Ali, M. S. A., & Ramli, N. (2020). A self-adapting ant colony optimization algorithm using fuzzy logic (ACOF) for combinatorial test suite generation. *IOP Conference Series: Materials Science and Engineering, 767*(1).
13. Rashid Ali, M. S. A., Othman, R. R., Yahya, Z. R., **Ahmad, M. Z. Z.**, & Ramli, N. (2016). Implementation of artificial bee colony algorithm for t-way testing. In *2016 3rd International Conference on Electronic Design (ICED)*, Phuket, Thailand.
14. Othman, R. R., **Ahmad, M. Z. Z.**, Rashid Ali, M. S. A., Zakaria, H. L., & Rahman, M. M. (2014). Effectiveness comparison of partially executed t-way test suite generated by existing strategies. In *International Conference on Mathematics, Engineering & Industrial Applications (ICoMEIA 2014)*, Penang, Malaysia.
15. Othman, R. R., Zamli, K. Z., Rashid Ali, M. S. A., & **Ahmad, M. Z. Z.** (2014). Input-input relationship constraints in t-way testing. In *1st UniMAP–PSU Engineering Postgraduate Conference*, Thailand.

16. Othman, R. R., Zamli, K. Z., Rashid Ali, M. S. A., & **Ahmad, M. Z. Z.** (2014). Addressing constraints support from within a t-way strategy. In *1st UniMAP–PSU Engineering Postgraduate Conference*, Thailand.

Accepted / In Press

1. Muhammad Aiman, M. A., **Ahmad, M. Z. Z.**, Othman, R. R., Halim, A. A. A., Go, K., & Al-Walidi, M. M. H. S. (2025). ISCSO: Input-output relation driven sand cat swarm optimization for combinatorial testing. In *Proceedings of the International Conference on Science, Engineering and Technology Application (ICOSETA 2025)*.
2. Murad, M. M. H. S. A.-W., Halim, A. A. A., Othman, R. R., **Ahmad, M. Z. Z.**, Go, K., & Asyraf, M. A. M. (2025). A variable strength interaction test suites generation strategy using sewing training optimization. In *Proceedings of the International Conference on Artificial Intelligence and Applied Sciences (i-CAIAS 2025)*.

CONFERENCE PARTICIPATION

2025 Presenter, International Exchange and Innovation Conference on Engineering & Sciences (IEICES 2025)

2023 Participant, International Symposium on Engineering and Technology (ISETech 2023)

2022 Presenter, International Conference on Software and Computer Applications (ICSCA 2022): **Excellent Paper Presentation**

2021 Presenter, 1st International Conference on Engineering and Technology (ICoEngTech 2021): **Best Paper Award**

2019 Presenter, International Symposium on Engineering and Technology (ISETech 2019)

AWARDS & RECOGNITION

2025 Anugerah Perkhidmatan Cemerlang, Anugerah Seri Gemilang UniMAP

2025 Anugerah Projek Komuniti (UniMAP for Society), Anugerah Kecemerlangan Penyelidikan

2025 Anugerah Pencetus Program 3U1I, Faculty of Electronic Engineering & Technology

2024 Anugerah Projek Komuniti (UniMAP for Society), Anugerah Kecemerlangan Penyelidikan

2022 Excellent Paper Presentation, ICSCA 2022: *VS-TACO: A Tuned Version of Ant Colony Optimization for Generating Variable Strength Interaction in T-Way Testing Strategy*

2021 Best Paper Award, ICoEngTech 2021: *A Tuned Version of Ant Colony Optimization Algorithm (TACO) for Uniform Strength T-way Test Suite Generator*

INNOVATION & EXHIBITION AWARDS

Award	Exhibition / Competition	Project
2026		
Gold	Education Innovation Challenge (EDIC), Transformative Teaching	*Transforming Course Administration: An Automated Management Ecosystem for the NMK31704 Design Project*
Silver	Education Innovation Challenge (EDIC), SULAM	*Hybrid TechShare Innovation Framework (HTIF): Hybrid SULAM Model for Enhancing STEM Engagement and Global Technology Knowledge Transfer*
Best Product Design	Integrated Design & Smart System Challenge (IDSSC 2026)	*Theme: Agricultural Application*
Silver	International Virtual Expo of Innovation Product and System Design (INVIDE 2026)	*Fuzzy-MedCare: Fuzzy Logic-Based Health Monitoring and Emergency Detection for Home Bedridden Patients*
Silver	International Virtual Expo of Innovation Product and System Design (INVIDE 2026)	*IoT-Based Smart Vehicle Anti-Theft System with Real-Time Monitoring*
Silver	AI & IoT International Innovation Expo (AIOTIE 2026)	*FuzzyFishCare: Design and Simulation of a Fuzzy Logic Decision Algorithm for Aquaculture Water Quality Control*
Gold	Perlis International Engineering Invention & Innovation Exhibition (Pi-ENVEX 2026)	*Fuzzy-MedCare: Fuzzy Logic-Based Health Monitoring and Emergency Detection for Home Bedridden Patients*

Award	Exhibition / Competition	Project
Gold	Perlis International Engineering Invention & Innovation Exhibition (Pi-ENVEX 2026)	*Smart Mushroom: AI-Driven Environmental Control System for Precision Mushroom Farming*
Silver	Perlis International Engineering Invention & Innovation Exhibition (Pi-ENVEX 2026)	*SandCatGo: An Intelligent Online Tool for Combinatorial T-Way Test Suite Generation Based on Sand Cat Swarm Optimization*
Gold	Postgraduate Colloquium & Research Exhibition 2026	*Introducing STBO for Efficient T-Way Test Suite Generation*
Gold	Ekspo Rekacipta UniMAP (EREKA 2026)	*Fuzzy-MedCare: Fuzzy Logic-Based Health Monitoring and Emergency Detection for Home Bedridden Patients*
Bronze	Ekspo Rekacipta UniMAP (EREKA 2026)	*SandCatGo: Intelligent T-Way Testing Platform*
Gold	Innovation Excellence (INNOVEX 2026)	*FlexiSolar: A Metaheuristic Testing Tool for Solar Dashboard Monitoring System*
Gold	Innovation Excellence (INNOVEX 2026)	*SandCatGo: Intelligent T-Way Testing Platform*
Gold	Innovation Excellence (INNOVEX 2026)	*Fuzzy-MedCare: Fuzzy Logic-Based Health Monitoring and Emergency Detection for Home Bedridden Patients*
Silver	Innovation Excellence (INNOVEX 2026)	*Self-Sustained Active Water-Cooling System for PV Modules using Irradiance-Based Control*
2025		
Gold	IDEA BESTARI Innovation Exhibition 2025	*Smart Fire Detection and Prevention System Using Fuzzy Logic*
Gold	IDEA BESTARI Innovation Exhibition 2025	*SandCatTest: Software Testing Tool Based on Sand Cat Swarm Optimization for Uniform T-Way Test Suite Generation*
Silver	IDEA BESTARI Innovation Exhibition 2025	*Fuzzy-MedCare: Fuzzy Logic-Based Health Monitoring and Emergency Detection for Home Bedridden Patients*
Gold	Ekspo Cipta 2025 (X-CIPTA 2025)	*SandCatTest: Software Testing Tool Based on Sand Cat Swarm Optimization for Uniform T-Way Test Suite Generation*
Gold	Ekspo Cipta 2025 (X-CIPTA 2025)	*Design and Development of T-Way Test Suite Generator Adopting Sewing Training-Based Optimization Algorithm*
Silver	Postgraduate Research Poster Competition, iFuture 2025 @ UniMAP	*Implementation of Sand Cat Swarm Optimization for Uniform T-Way Test Suite Generation*
Silver	Sulam Innovate Showcase and Symposium (SUISS 2025)	*Hari Teknologi & Masyarakat 2024 (HTM2024): Empowering Youth Through STEM Activities with Rumah Kanak-Kanak Arau*
Gold	Ekspo Rekacipta UniMAP (EREKA 2025)	*Adaptive Water-Cooling System for Photovoltaic Modules Using AI Optimization*
Gold	Ekspo Rekacipta UniMAP (EREKA 2025)	*IoT-Based Smart Paddy Field Monitoring Crop Management System*
Silver	Ekspo Rekacipta UniMAP (EREKA 2025)	*Intelligent Parcel Delivery and Notification System for Apartment House*
Silver	Ekspo Rekacipta UniMAP (EREKA 2025)	*Smart Waste Management System with IoT Integration using ESP32*
Silver	AI & IoT International Innovation Expo (AIOTIE 2025)	*Intelligent Parcel Delivery and Notification System for Apartment House*
Gold	Integrated Design Project Exhibition (IDPEx 2025)	*UniTAG: Renewable Vehicle Access System for UniMAP Campus using RFID*
Participant	Pertandingan SULAM Innovate Ke-3, 3rd International Conference on SULAM 2025	–
2024		
Silver	Ekspo Rekacipta UniMAP (EREKA 2024)	*IoT Warning System for Early Forest Fire Detection*

Award	Exhibition / Competition	Project
Bronze	AI & IoT International Innovation Expo (AIOTIE 2024)	*IoT Warning System for Early Forest Fire Detection*
Participant	Undergraduate Project Exhibition (UPEx 2024), FKTEN	–
2020		
Gold	Virtual Research and Innovation Exhibition (EREKA 2020)	*TTSGA: T-Way Test Suite Generator*
Gold	International Invention, Innovation & Technology Exhibition (ITEX 2020)	*TTSGA: T-Way Test Suite Generator*
2015		
Gold	International Engineering Invention and Innovation Exhibition (I-ENVEX 2015)	*GVS: T-Way Test Suite Generator Supporting Constraints*
Silver	Malaysia Technology Expo (MTE 2015)	*GVS: T-Way Test Suite Generator Supporting Constraints*
Silver	International Invention, Innovation & Technology Exhibition (ITEX 2015)	*GVS: T-Way Test Suite Generator Supporting Constraints*

TEACHING EXPERIENCE

No.	Course	Role	Semester
Bachelor of Electronic Engineering Technology (Electronic Systems) with Honours			
1.	NMK20603 – Computer Architecture	Team Teaching	Semester 2, 2025/2026
2.	NMK31704 – Design Project	Course Coordinator	Semester 2, 2025/2026
3.	NMK42003 – Instrumentation	Course Coordinator	Semester 1, 2025/2026
4.	NMK10702 – Writing in Engineering Technology	Course Coordinator	Semester 1, 2025/2026
5.	NMK31704 – Design Project	Course Coordinator	Semester 2, 2024/2025
6.	NMK10803 – Digital Systems	Team Teaching	Semester 2, 2024/2025
7.	NMK42003 – Instrumentation	Course Coordinator	Semester 1, 2024/2025
8.	NMK20603 – Computer Architecture	Team Teaching	Additional Semester, 2024/2025
9.	NMK31703 – Design Project	Course Coordinator	Semester 2, 2023/2024
10.	NMK42003 – Instrumentation	Course Coordinator	Semester 1, 2023/2024
11.	NMK10603 – C Programming	Team Teaching	Semester 2, 2022/2023
12.	PGT106 – C Programming	Course Coordinator	Semester 2, 2022/2023
13.	NMK10603 – C Programming	Course Coordinator	Semester 1, 2022/2023
14.	PGT106 – C Programming	Course Coordinator	Semester 1, 2022/2023
Diploma in Computer Engineering			
1.	DKT223 – Database Systems	Course Coordinator	Semester 2, 2017/2018
2.	DKT121 – Fundamentals of Computer Programming	Course Coordinator	Semester 2, 2017/2018
3.	DKT216 – Object-Oriented Programming	Course Coordinator	Semester 1, 2017/2018
4.	DKT121 – Fundamentals of Computer Programming	Course Coordinator	Semester 2, 2016/2017
5.	DKT311 – Software Engineering	Course Coordinator	Semester 1, 2016/2017

SUPERVISION

Postgraduate (PhD)

Student	Thesis Title	Role / Status
Muhammad Aiman bin Mohd Asyraf PhD (Computer Engineering), Full-time Research	A New Phase Transition Controller in Sand Cat Swarm Optimization for Combinatorial Test Suite Generation	Co-Supervisor Ongoing
Siti Shamirah binti Sabri PhD (Computer Engineering), Full-time Research	Design and Development of a Multi-factorial Sand Cat Swarm Optimization Algorithm for T-Way Combinatorial Testing	Co-Supervisor Ongoing

Final Year Project (Undergraduate)

No.	Student (Matric No.)	Session	Project Title
Bachelor of Electronic Engineering Technology (Electronic Systems) with Honours			
1.	Muhammad Sufian Hakeem bin Sofi (231373117)	2025/2026	FuzzyDamGuard: Fuzzy Logic-Based Dam Monitoring and Gate Control using ESP32
2.	Vaashienie a/p Santhiramogan (231373903)	2025/2026	FuzzyFishCare: ESP32-Based Intelligent Aquaculture Monitoring and Aeration Control Using Fuzzy Logic
3.	Nurul Alya binti Abdul Aziz (221373175)	2025/2026	Fuzzy-AirSense: In-Car Air Quality Monitoring & Safety System using Fuzzy Logic
4.	Muhammad Fadlan Adli bin Muhamad Amin (221372354)	2025/2026	Fuzzy-AWOS: AI-Driven Weather & Runway Safety Monitoring System for Airports
5.	Mohanambal a/p Loganathan (221373148)	2025/2026	Fuzzy-MedCare: Fuzzy Logic-Based Health Monitoring and Emergency Detection for Home Bedridden Patients
6.	Nor Amirah Syazwani binti Kamarudin (221372357)	2024/2025	Smart Office System: Resources, Security, Environmental Control
7.	Thejashviny a/p Sathgunalen (211372782)	2024/2025	IoT Weather Station with Early Flood Detection System in Urban Areas
8.	Sharwini a/p Mugunthan (211371439)	2024/2025	Intelligent Parcel Delivery and Notification System for Apartment Houses
9.	Muhammad Haikal bin Hamdi (201391576)	2023/2024	IoT Warning System for Early Forest Fire Detection
10.	Sharul Haqimi bin Md. Halim (191383231)	2022/2023	Automatic Cat Feeder and Litter Cleaner with Android Application
Bachelor of Engineering (Hons) Computer Engineering			
1.	Nur Farah Hanim binti Mat Ali (141020924)	2017/2018	Online UniMAP's Qurban Management System
2.	Che Muhamad Amin bin Che Jaafar (141020865)	2017/2018	Mobile Queue Management System
3.	Tan Chang Cun (141020947)	2017/2018	Online System for Property Sales and Management
4.	Hilal Sheikh Ali Nur (141020086-1)	2017/2018	Online Voting System for UniMAP's Election Management

Design Project (Group Supervision)

No.	Students (Matric No.)	Session	Project Title
Bachelor of Electronic Engineering Technology (Electronic Systems) with Honours			
1.	Muhammad Nazri bin Kasim (231393669) Darawadi Khewmani a/p Su Tin (231373096) Ahmad Irfan Darwisy bin Shahrizan (221303003) Muhammad Izad bin Ahmad Zafri (241302502)	2025/2026	Development of an Automated Integrated Aquaculture-Agriculture (IAA) System for the Asnaf Community

No.	Students (Matric No.)	Session	Project Title
2.	Mohamad Zariff Daniel bin Zulcalfle (221301700) Mohamad Sallehudin bin Ismail (221391680) Tengku Ahmad Naqiuddin bin Tg Mohd Amran (221373185) Muhammad Afnan bin Abdullah (231302490)	2024/2025	UniTAG: Renewable Vehicle Access System for UniMAP Campus Using RFID
3.	Muhammad Nabil Aiman bin Nor Asrol (211391454) Zarul Al-Amin bin Zamri (211302824) Priatharsshen a/l Suntharalingam (211372771) Nur Amalina binti Abu Seman (211392806) Diviyaa a/p Muniandy (211373353)	2023/2024	Safeguarding Apartments: IoT Fire Detection and Prevention System

UNIVERSITY SERVICE & LEADERSHIP

Leadership & Coordination

2026 Google AI Champion, Universiti Malaysia Perlis

2026 Coordinator, HYBRID TECHSHARE: UniMAP–Yala Global SULAM 2026, Service Learning Malaysia – University for Society (SULAM)

2025/26 Coordinator, Work-Based Learning (WBL) Course (3u1i), Bachelor of Electronic Engineering Technology (Electronic Systems)

2025 Deputy Director, Asia Campus Plus – Short Mobility Program 2025, UniMAP

2025 Coordinator, STEM Programme "Build-It Buddies: FKTEN – Sekolah Rendah Islam III" (SULAM)

2025 Team Manager, Woodball (Women), FKTEN, Karnival Sukan UniMAP 2025

2024–26 Leader, Big Data & Cloud Computing Cluster, Centre of Excellence for Advanced Computing (AdvComp)

2024 Coordinator, Program Hari Teknologi dan Masyarakat 2024 (SULAM)

2024 Coordinator, "Lawatan Rumah Kanak-Kanak Arau ke UniMAP – Eksplorasi UniMAP: Kembara Sains, Teknologi dan Media FKTEN" (SULAM)

2023 Team Manager, FKTEN, Karnival Sukan UniMAP (KSU) 2023

2022–26 Research Fellow, Centre of Excellence for Advanced Computing (AdvComp)

2022 President, Muallaf Care Team, One Centre Malaysia Perlis

2019 Team Manager, PPKKP, Karnival Sukan UniMAP (KSU) 2019

2018 Team Manager, PPKKP, Karnival Sukan UniMAP (KSU) 2018

2018 Deputy Director, Program Ukhuwah PPKKP bersama Anak Yatim

2017 Deputy Team Manager, PPKKP, Karnival Sukan UniMAP (KSU) 2017

Committee Appointments

2026 Facilitator, "Future IC Design Engineer and STEM Carnival 2026" (MiCTEC UniMAP, MOSTI STEM Camp Grant)

2026 Program Committee, International Engineering Postgraduate Colloquium (IEPC 2026)

2026 Judging Committee, Integrated Design & Smart System Challenge (IDSSC 2026)

2025–26 Committee, KPI Multimedia, FKTEN

2023–26 Committee, KPI Kerohanian & Kebajikan Raudhah, FKTEN

2025 Committee, Program Pemerkasaan Pengaturcaraan Komputer Pelajar Tingkatan 6 STPM (also 2022, 2023)

2024 Logistics and Transportation Committee, Asia Campus Plus – Short Mobility Program 2024, UniMAP

2024 Committee, Engineering Camp UniMAP–RUTS 2024 (Smart Farm Camp), Thailand

2024 Moderator, Tazkirah Raudhah@FKTEN

2023 Local Arrangement & Logistics Committee, International Symposium on Engineering and Technology (ISETech 2023)

2023 Technical Committee, AdvComp 3rd Workshop Series: "Mendeley Reference Manager & Systematic Literature Review"

2023 Technical Committee, AdvComp 5th Workshop Series: "User Behavioral Research on Human Computer Interaction"

2023 Committee, Karnival Mini Usahawan Asnaf Perlis (IIUM & AdvComp UniMAP)
2022 Committee, Kolej Vokasional Seberang Perai to UniMAP Benchmarking Visit
2019 Committee, SCCE Postgraduate and Research Seminar Series
2018 Secretariat, Innovation, Project Presentation and Exhibition (IPRoPEX)
2018 Secretariat, "Program Minggu STEM & Kerjaya"
2018 Secretariat, Karnival Pendidikan Tinggi Negara (KPTN) 2018, Zon Mini Perlis dan Kedah
2018 Committee, Promosi Universiti Malaysia Perlis
2017 Committee, Promosi School of Computer and Communication Engineering
2017 Committee, Pelaksana MPP 2017
2017 Committee, Santapan Malam Konvokesyen 2017 (Physical & Logistics Bureau; Security Bureau)

Academic Assessment

2026 Industrial Training Evaluation Panel (PPLI), NMK40412, Semester 2, 2025/2026
2026 Panel Evaluator, NMK31704 Design Project
2025 Panel Evaluator, NMK31704 Design Project
2024 Panel Evaluator, NMK31703 Design Project
2024 Industrial Training Evaluation Panel, Semester 2, 2023/2024
2024 Internal Panel Examiner, Pre-Viva, M.Sc. (Computer Engineering), Candidate: Mohd Mizan bin Munif, FKTEN
2023 Industrial Training Evaluation Panel, Semester 2, 2022/2023
2018 Industrial Training Evaluation Panel, Semester 2, 2017/2018
2018 Panel Evaluator, Diploma Exhibition on Technology and Engineering (DEXTEr)

Reviewer

2026 IEEE International Conference of Electrical, Electronic and Computer Engineering (ICEECE)
2025 9th International Conference on Man-Machine Systems (ICoMMS)
2025 eBook Juadah Minda Peringkat Kebangsaan (Politeknik Port Dickson)
2024 International Symposium on Microelectronic Engineering & Design (ISMED)
2024 7th International Conference on Electronic Design (ICED)
2024 5th International Conference on Mathematics, Engineering & Industrial Applications (ICoMEIA)
2023 3rd International Symposium on Engineering and Technology (ISETech)
2023 2nd International Conference on Engineering and Technology (ICoEngTech)
2022 6th International Conference on Electronic Design (ICED)

Judge / Jury

2026 Judge, AI & IoT International Innovation Expo (AIOTIE 2026)
2026 Jury, International Virtual Expo of Innovation Product and System Design (INVIDE 2026)
2025 Judge, AI & IoT International Innovation Expo (AIOTIE 2025)
2025 Jury, Ekspo Cipta 2025 (X-CIPTA 2025)
2024 Judge, Karnival STEM Peringkat Negeri Perlis
2024 Judge, AI & IoT International Innovation Expo (AIOTIE 2024)
2024 Judge, Exhibition for Young Agricultural & Food Engineers and Technologies

INVITED SPEAKER & WORKSHOPS

2026 "Innovation in Education". Organizer: Pusat Pembangunan Kepimpinan Pendidikan Lestari (LEAD), UniMAP
2026 "My Professional URL: Transforming Traditional CVs into Digital Portfolios" Workshop. Organizer: Centre of Excellence for Advanced Computing (AdvComp), UniMAP
2026 "Lensa Sains: Menyingkap Rahsia Al-Quran di Era Teknologi". Organizer: Tham Suksa Islamic School, Yala, Thailand
2026 "Sistem dan Pelaksanaan Projek Tahun Akhir" briefing, invited lecturer for Bachelor of Industrial Electronic Automation Technology (UR6523011). Organizer: Department of Applied Electronics, FKTEN, UniMAP

2025 "Hybrid Metaheuristic Approaches for T-Way Test Suite Generation". Organizer: University of Yamanashi, Japan

2025 "MPKK IRC Workshop: Paperwork Writing Using AI Tools & Semester 2 Strategic Planning". Organizer: International Resident College & Non-Resident (IRCNR), UniMAP

INTERNATIONAL MOBILITY

2026 Inbound: Program Mobiliti Antarabangsa IRCNR UniMAP bersama Universitas Islam Negeri (UIN) Ar-Raniry Banda Aceh, Indonesia

2025 Outbound: Research Collaboration Visit, University of Yamanashi, Japan

2024–25 Inbound: A3I Campus Asia Plus Program

2024 Outbound: Engineering Camp UniMAP–RUTS 2024, Thailand

PROFESSIONAL CERTIFICATIONS

2026 Gemini Certified Faculty, Google for Education

2012 IBM Certified System Administrator – WebSphere Application Server Network Deployment V6.1

2012 IBM e-Business Administrator

2009 Certified Fiber Optic Technician (CFOT)

TRAINING & COURSES ATTENDED

2026 Kursus Pensijilan Kompetensi Pengajar Pendidikan, Latihan Teknikal & Vokasional (TVET) UniMAP, LEAD UniMAP (3–6 Aug 2026)

2026 Google for Education Gemini Academy Workshop

2026 3D Design and Print (FreeCAD + OrcaSlicer)

2026 Webinar: Work-Based Learning (WBL)

2024 Workshop: Fundamentals of Multifactorial Optimization, Metaheuristics and Hyper-heuristics

2024 Kursus Asas Google Sites

2022 MyRESKILL IoT Programme

2022 Google Certified Educator Level 1 Training

2022 Kursus Pengurusan Kewangan Berhemah

2022 Bengkel Pengurusan Muallaf

2019 LabVIEW Workshop

2018 Deploying Hyper-heuristic & Metaheuristic Algorithms Workshop (UMP)

2018 IEEE Distinguished Lecturer Program

2018 Kursus Pengurusan Jenazah

2018 Bengkel Bio Terapi Solat

2017 Kursus Asas Pengajaran dan Pembelajaran (PdP)

2017 Program Transformasi Minda

2013 i5 and System i System Operator Workshop (AS24MY), IBM

2013 Advanced System Operator Workshop for IBM i-Series

REFERENCES

Assoc. Prof. Dr. Rozmie Razif bin Othman
Faculty of Intelligent Computing (FKC)
Universiti Malaysia Perlis (UniMAP)
Email: rozmie@unimap.edu.my

Prof. Ir. Ts. Dr. R. Badlishah Ahmad
Faculty of Intelligent Computing (FKC)
Universiti Malaysia Perlis (UniMAP)
Email: badli@unimap.edu.my